

Name : Vivek Chaudhari

Project : 10

NodeDock : Containerized Node.js application deployed on AWS



Project Overview

NodeDock is a cloud-based deployment project where a Node.js application is containerized using Docker and deployed on AWS. The Docker image is stored in a private registry and executed using a managed container orchestration service, ensuring consistent, scalable, and production-ready deployment.

Working Flow (End-to-End)

1. Developer builds a Node.js application (API or web app)
2. Application is containerized using Docker
3. Docker image is pushed to a container registry
4. Container service pulls the image
5. Application runs as containers in the cloud
6. Users access the application via endpoint

⚙️ AWS Services Used & Their Role

📦 Amazon Elastic Container Registry

- Stores Docker images securely
- Maintains versioned images for deployment

- Acts as source for container runtime
-

Amazon Elastic Container Service

- Deploys and manages containers
 - Handles scaling and availability
 - Runs application without manual server management
-

Core Technologies

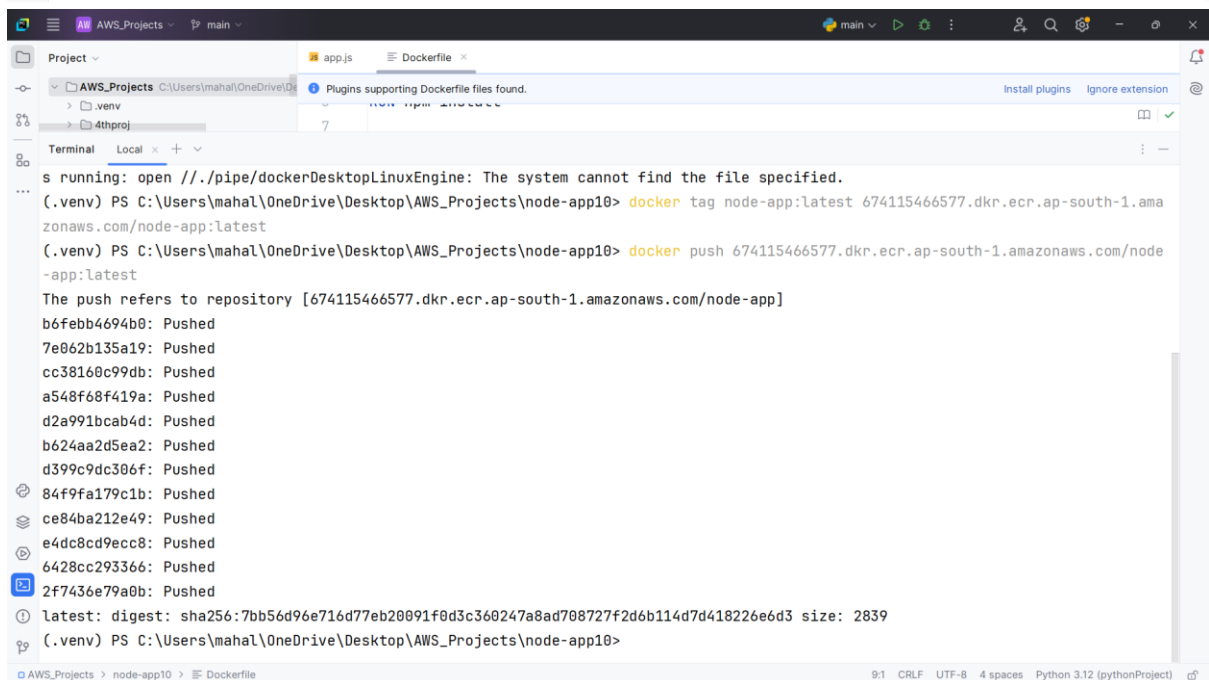
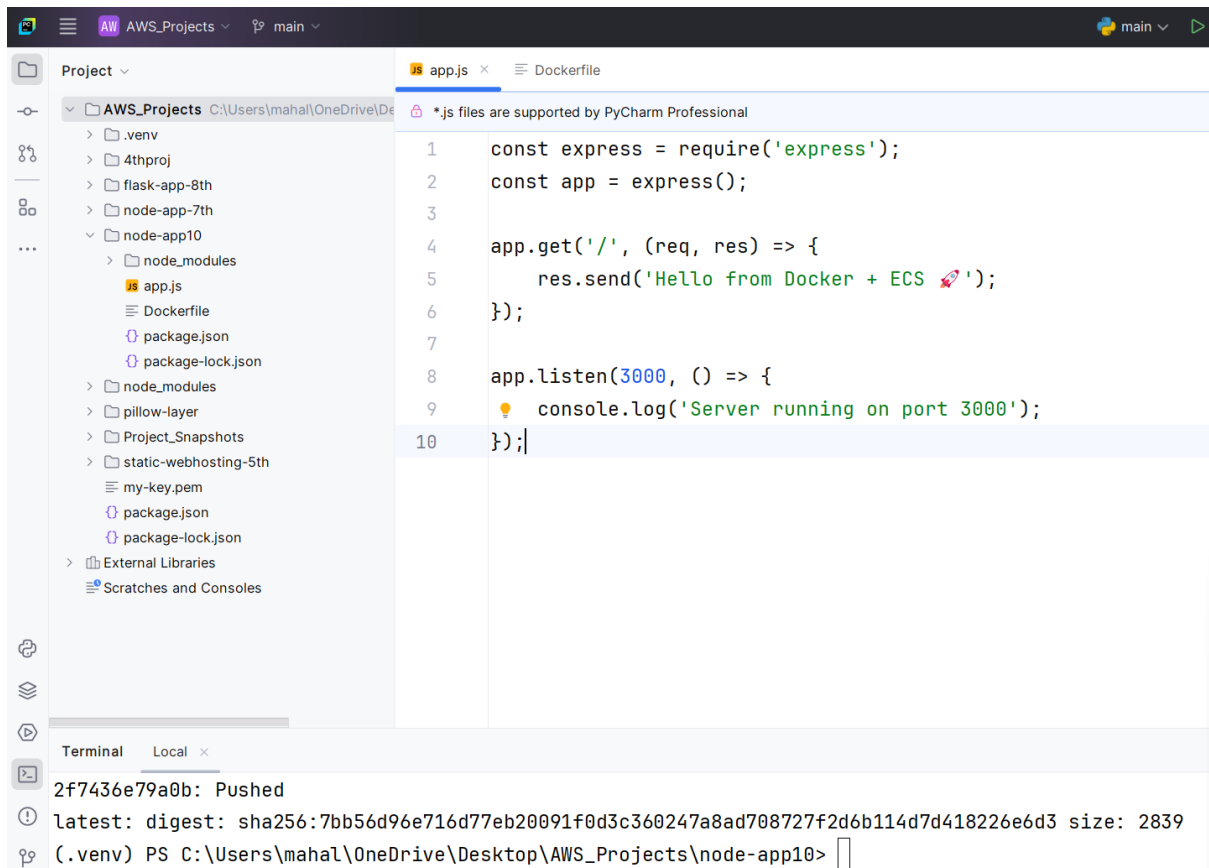
- **Node.js** → Backend runtime
 - **Docker** → Containerization
 - **AWS ECS & ECR** → Deployment & orchestration
-

Key Objectives Achieved

- ✓ Consistent deployment across environments
 - ✓ Scalable container-based architecture
 - ✓ Simplified deployment using Docker
 - ✓ Reduced dependency conflicts
-

Business / Real-World Value

- Faster and reliable deployments
- Easy scaling based on traffic
- Reduced operational complexity
- Production-ready DevOps workflow



Hello from Docker + ECS

Amazon Elastic Container Service

Express Mode

Clusters

Namespaces

Task definitions

Daemon task definitions [New](#)

Account settings

AWS Batch [i](#)

Amazon ECR [i](#)

Repositories [i](#)

Online learning workshop [i](#)

Documentation [i](#)

Discover products [i](#)

Subscriptions [i](#)

Tell us what you think

Amazon Elastic Container Service

Clusters > node-cluster > Services > node-task-service-22a79zdg > Tasks > 9602ca290c744ad0b86fa27fdd3f7f7 > Configuration

Configuration Metrics Logs Networking Volumes (0) Tags

Task overview

ARN
arn:aws:ecs:ap-south-1:674115466577:task/node-cluster/9602ca290c744ad0b86fa27fdd3f7f7

Last status
Running

Desired status
Running

Started/Created at
April 26, 2026, 00:48 (UTC+5:30)
April 26, 2026, 00:48 (UTC+5:30)

Started by
ecs-svc/0230055615601080959

Fargate ephemeral storage

Encryption
Default AWS Fargate encryption

Size (GiB)
20

Configuration

Operating system/Architecture
Linux/x86_64

Capacity provider
Fargate

ENI ID
eni-01a4ae6bc1eca85e9 [i](#)

Public IP
3.110.67.242 | [open address](#) [i](#)

CPU | Memory
25 vCPU | 5 GiB

Launch type
Fargate

Network mode
awsvpc

Private IP
172.31.17.237

Amazon Elastic Container Service

Task definitions > node-task

node-task-service-22a79zdg has been deployed successfully. [View service](#) [x](#)

node-task (1) [Info](#) Last updated April 26, 2026, 01:03 (UTC+5:30) [Deploy](#) [Actions](#) [Create new revision](#)

☐ Task definition: revision

☐ node-task-1 Active

Amazon ECR

Private registry

Repositories

node-app

sha256:7bb56d96e716d77eb20091f0d3c360247a8ad708727f2d6b114d7d418226e6d3

hitesh_r_mahale

🔍

🔔

Amazon Elastic Container Service

Private registry

Repositories

node-app

Features & Settings

Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

Getting started

Documentation

Image

Archive

Details

Image tags

latest

URI

674115466577.dkr.ecr.ap-south-1.amazonaws.com/node-applatest

Digest

sha256:7bb56d96e716d77eb20091f0d3c360247a8ad708727f2d6b114d7d418226e6d3

General information

Artifact type

Image

Repository

node-app

Pushed at

April 25, 2026, 23:44:07 (UTC+05.5)

Last recorded pull time

April 26, 2026, 00:41:42 (UTC+05.5)

Size (MB)

398.30

Image status

Active

Scanning and vulnerabilities

Status

Scan not found

Scan

Amazon Elastic Container Service

Task definitions

node-task

Revision 1

Containers

node-task-service-22a79cdg has been deployed successfully.

View service

Last updated

April 26, 2026, 01:04 (UTC+5.30)

Deploy

Actions

Create new revision

Amazon Elastic Container Service

Express Mode

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Daemon task definitions

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Tell us what you think

node-task:1

Overview

ARN

arn:aws:ecs:ap-south-1:674115466577:task-definition/node-task:1

Status

ACTIVE

Time created

April 26, 2026, 00:01 (UTC+5.30)

App environment

Fargate

Task role

-

Task execution role

ecsTaskExecutionRole

Operating system/Architecture

Linux/X86_64

Network mode

awsvpc

Fault injection

turned off

Containers

JSON

Task placement

Volumes (0)

Requires attributes

Tags

Task size

Task CPU

256 units (0.25 vCPU)

Task CPU maximum allocation for containers

Task memory

512 MiB (0.5 GiB)

Task memory maximum allocation for container memory reservation

Gordon

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit

Docker Hub

Docker Scout

Extensions

Containers / kind_driscoll

kind_driscoll

982bcce2a864

node-app:latest

STATUS

Running (0 seconds ago)

Logs

Inspect

Bind mounts

Exec

Files

Stats

Server running on port 3000

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit

Docker Hub

Docker Scout

Extensions

Local

My Hub

2.35 GB / 1.25 GB in use

5 Images

Last refresh: 4 hours ago

Search

Delete

Space to be reclaimed 1.09 GB

	Name	Tag	Image ID	Created	Size	Actions
☒	node-app	latest	2d231940b5bc	4 hours ago	1.09 GB	
☐	<none>	<none>	bc15d7916a72	1 day ago	1.1 GB	
☐	flask-ecs-app	latest	aa840de423c	5 days ago	132.52 MB	
☐	tictactoe	latest	91d570c45823	14 days ago	1.1 GB	
☐	myapp	latest	a0d21a2c72bc	14 days ago	1.1 GB	

